Frequency-Domain Guided Swin Transformer and Global–Local Feature Integration for Remote Sensing Images Semantic Segmentation

Haoxue Zhang, Gang Xie, *Member, IEEE*, Linjuan Li, Xinlin Xie, and Jinchang Ren, *Senior Member, IEEE*

***Abstract*— Convolutional neural networks (CNNs), transformers, and the hybrid methods have been significant application in remote sensing. However, existing methods are limited in effectively modeling frequency-domain information, which affects their ability to capture detailed information. Therefore, we propose a frequency-domain guided feature coupled mechanism and a global–local feature integration method (FGNet) for semantic segmentation. Specifically, a frequency-domain guided Swin (FGSwin) transformer is designed by introducing dilation group convolution, fast Fourier transform (FFT), and learnable weights to enhance the expression capability of frequency-domain and space-domain, local and global features, simultaneously. In addition, a global–local feature integration (GLFI) module is proposed for aggregating features to further enhance the discrimination of each category. Comprehensive experimental results demonstrate that compared with existing methods, the proposed method achieves superior performance in terms of mean intersection over union (mIoU), reaching 71.46% and 74.04% on ISPRS Potsdam and Vaihingen, two widely used datasets.**

***Index Terms*— Feature integration, frequency-domain guided Swin (FGSwin) transformer, global–local features, remote sensing semantic segmentation.**

I. Introduction

With the development of remote sensing technology, the availability of high-resolution images has increased, and the research on remote sensing semantic segmentation methods has been intensified. The purpose of semantic segmentation is to assign labels for each pixel to obtain the interpretation of the whole image, which has important applications in precision agriculture [1], urban development [2], natural resource utilization [3], and disaster assessment [4].

Semantic segmentation of remote sensing images consists of traditional learning-based methods and deep-learning-based methods, the former is accomplished by manually designing features and segmentation criteria, whose generalization ability and robustness need to be strengthened; the latter automatically captures the image information with a high level of intelligence and has received extensive attention in recent years. Among the methods based on deep learning, those of convolutional neural networks (CNNs), transformers, and their hybrid architectures occupy the mainstream.

CNNs extract local image features through convolution operations, enabling multiscale feature representation. Fully convolutional networks (FCNs) [5] introduced per-pixel classification, followed by U-net [6] with a symmetric encoder–decoder structure. PSPNet [7] and FPN [8] expanded receptive fields with spatial pyramid pooling, while other methods [9], such as orientation attention network [10] and stair fusion network [11], incorporated attention mechanisms. However, CNNs rely on local operations, limiting their ability to capture global information, which is critical for semantic segmentation. To address the limitations of CNNs in modeling global context, transformers use a multihead self-attention (MSA) mechanism to extract global information, overcoming the constraints of local receptive fields in convolutional networks. This capability enhances image interpretation by differentiating objects of the same category from others [12], [13]. The Rssformer [14] uses an adaptive transformer fusion module to suppress background noise and enhance foreground saliency. The boundary-aware multiscale network [15] introduces a scale attention module to construct long-range dependencies. Mixed-mask transformer [16] uses a hierarchical encoder for multiscale learning. Despite their effectiveness in capturing global information, transformers face challenges in computational complexity. The Swin transformer [17] mitigates this issue through nonoverlapping local windows, ensuring linear scalability with image size. Nonetheless, further optimization of information extraction mechanisms is necessary, highlighting the potential of hybrid architectures.

Recognizing the strengths of CNNs in capturing local information and the capability of transformers to model global features, hybrid architectures have been developed to integrate the benefits of both the approaches. UNetFormer [18] and CNNs and multiscale transformer fusion network (CMTFNet)

Received 26 November 2024; revised 6 January 2025; accepted 24 January 2025. Date of publication 28 January 2025; date of current version 28 February 2025. This work was supported in part by the Fundamental Research Funds for the Key Research and Development Plan of Shanxi Province under Grant 202202010101005, in part by the Industry-University-Research Innovation Fund for Chinese Universities under Grant 2021ZYA11005, in part by Guangdong Province Key Construction Discipline Scientific Research Ability Promotion Project under Grant 2022ZDJS015 and Grant 2021ZDJS025, in part by the Special Projects in Key Fields of Ordinary Universities of Guangdong Province under Grant 2021ZDZX1087, and in part by Guangzhou Science and Technology Plan Project under Grant 2024B03J1361 and Grant 2023B03J1327. *(Corresponding author: Gang Xie.)*

Haoxue Zhang, Gang Xie, Linjuan Li, and Xinlin Xie are with Shanxi Key Laboratory of Advanced Control and Industrial Intelligence and the School of Electronic and Information Engineering, Taiyuan University of Science and Technology, Taiyuan 030024, China (e-mail: zhanghaoxue95@stu.tyust.edu.cn; xiegang@tyust.edu.cn; linjuanli@tyust.edu.cn; xiexinlin@tyust.edu.cn).

Jinchang Ren is with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou 510640, China, and also with the School of Computing, Engineering and Technology, Robert Gordon University, AB10 7AQ Aberdeen, U.K. (e-mail: jinchang.ren@ieee.org).

Digital Object Identifier 10.1109/TGRS.2025.3535724

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

[19] build a hybrid architecture, which use CNNs as an encoder and transformer as a decoder. Conversely, class-guidance network [20] uses a transformer as an encoder to mine the class-specific perceptual information for each semantic class, while CNNs as a decoder to enhance the information. Moreover, SRCBTFusion-Net [21] is designed through a cascade dual-coding structure and a multiscale upsampling integration module in which resolution information can be maintained. But the cascade structure usually increases the number of parameters, which requires the use of lightweight backbone network of pretrained for feature extraction. Therefore, dual-branch networks, due to their efficiency, have been gradually proposed, such as STUNet [22], a global and edge enhanced transformer [23], and CSTUNet [24].

While hybrid architectures combining CNNs and transformers effectively integrate spatial-domain information, they predominantly emphasize spatial representations, often neglecting the potential of frequency-domain modeling. In contrast, robust representation of frequency-domain information significantly enhances semantic segmentation performance, as high-frequency characteristics—particularly lines and edges—play a crucial role in delineating category boundaries and improving overall accuracy (OA). A straightforward approach is to incorporate the frequency domain into convolution or attention mechanisms. For instance, frequency-adaptive dilated convolution [25] dynamically adjusts dilation rates based on local frequency components, whereas frequency-domain feature-guided network [26] uses a frequency enhancement attention module to identify and strengthen frequency details. While these methods have made some progress in feature extraction, there is still scope for improvement in the effective utilization of frequency characteristics, which may restrict the performance of the models in complex scenarios. To make more effective use of the frequency-domain features, multiscale frequency attention gating network [27] and SFFNet [28] focus on frequency decomposition, which isolates image components across various frequency bands, particularly around edges. In addition, the frequency-driven edge network [29] sharpens boundary definitions using a 2-D discrete wavelet transform to suppress low-frequency noise while accentuating salient edge details within spatial features. SSCNet [30] proposes spectral attention to capture the spectral context in the frequency domain, which maps the feature map into the frequency domain and calculates edge loss for high-frequency components. Despite the effectiveness of these methods in addressing complex targets, they still require refinement in their integration with spatial information to enhance both robustness and precision. A growing number of approaches now aim to explore more sophisticated frequency–space fusion strategies. MsanlfNet [31] combines multiscale attention with a nonlocal filter to process spatial and frequency characteristics at multiple scales. MIFNet [32] builds on this by incorporating local, global, and frequency data within a unified module. The dual-domain fusion network [33], based on wavelet frequency decomposition and fuzzy spatial constraints, further enhances segmentation by integrating spatial and frequency information. Wavelet feature enhancement [34] performs a multiscale, lossless decomposition of the input image, which helps preserve high-frequency details. Spatial–frequency network [35] embeds contextual feature dependencies in both the spatial and frequency domains. Nevertheless, current approaches typically incorporate frequency-domain transformations within convolutional or attention mechanisms or integrate frequency-domain information as a supplementary module. However, these strategies exhibit inherent limitations, as any information lost during the feature extraction process cannot be effectively recovered in later stages. Thus, there is a pressing need for further research to focus on optimizing the interplay between frequency-domain and spatial-domain processing during feature extraction.

Therefore, based on the hybrid architecture, we propose a frequency-domain information enhancement mechanism, which is integrated into the feature extraction process to improve the combined representation of global–local and frequency–space-domain information. In addition, we design a global–local information integration method to further enhance feature effectiveness. Our main contributions are summarized as follows.

1) Using learnable weights and shallow feature extraction methods, we propose a frequency-domain information enhancement mechanism (FGSwin) and couple it into Swin transformers, which can model frequency-domain and space-domain information and is different from the existing frequency information using approaches. By improving the feature description capability and coupling it into the Swin transformer for better long-range-dependent information modeling.
2) We develop a global–local feature integration (GLFI) module using a small convolutional kernel and a more easily convergent activation function. This design enhances class discrimination while maintaining a low parameter count in the network.
3) A hybrid architecture for semantic segmentation of remote sensing images based on FGSwin, residual CNNs, and GLFI is designed. This architecture achieves performance of 71.46% and 74.04% on the widely used Potsdam and Vaihingen datasets, which demonstrates competitive results compared with the state-of-the-art (SOTA) methods.

The remainder of this article is organized as follows. Section II provides the motivation and overview of the FGNet. In Section III, the results of specific experiments are reported. In Section V, the conclusions are summarized.

II. Methodology

A. Motivation and Overall Architecture of the Network

Semantic segmentation is a process of recognizing homogeneity and heterogeneity information in the image; usually, homogeneity manifests itself in low frequency, and heterogeneity can be distinguished by high-frequency information. Frequency-domain analysis converts an image from the physical space into the frequency space with clear meaning. It can not only obviously distinguish between low-frequency components and high-frequency components but also reduce processing complexity. Therefore, the methods modeling frequency characteristics can improve the performance of semantic segmentation theoretically [27], [36], [37].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

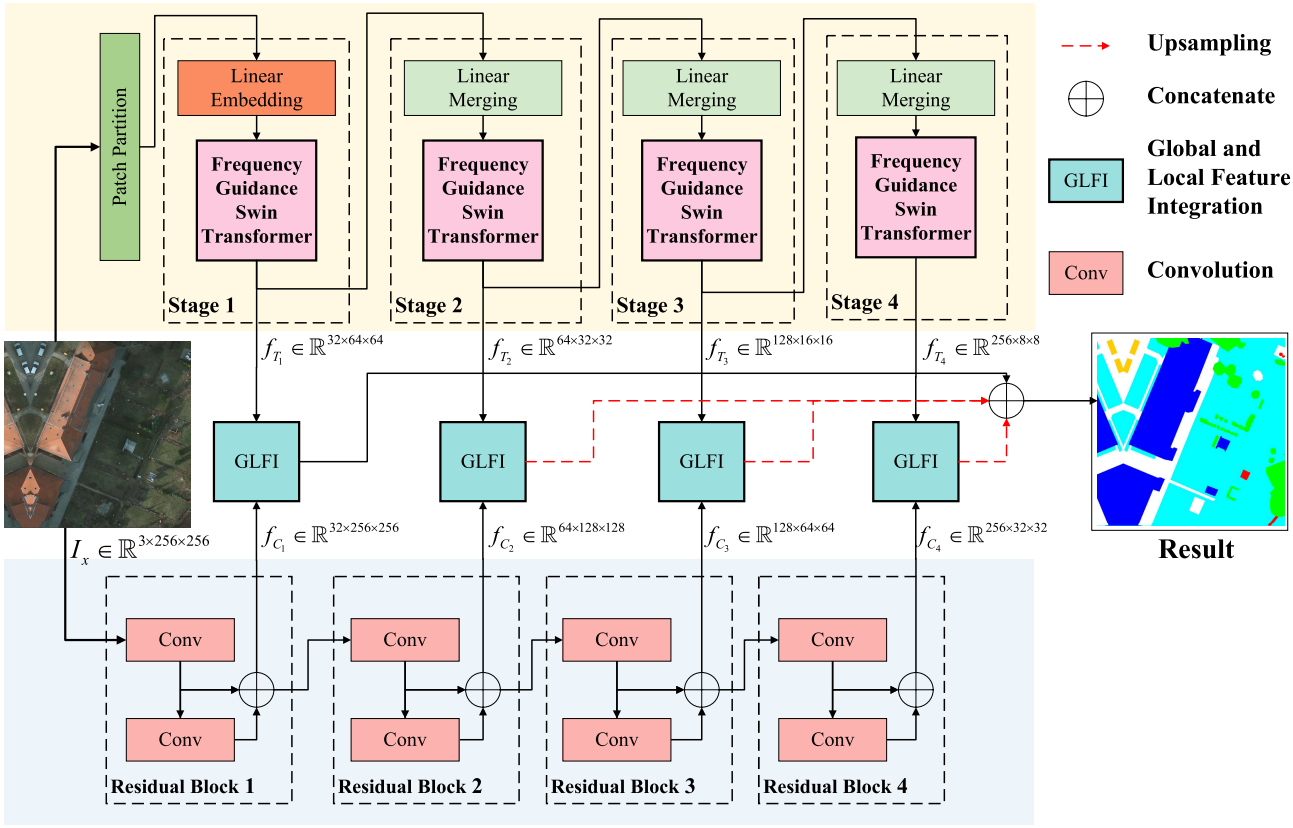


Fig. 1. Overview of the proposed FGNet.

However, the existing CNN-based and transformer-based methods only capture local and global information in the physical space and are limited in analyzing frequency features. The existing frequency analysis methods usually represent the frequency information by constructing the fusion module or attention in combination with the network [29], [31], [37] and the validation of more effective coupling ways and applications in semantic segmentation need to be further investigated. Thus, we propose a frequency-domain guided feature enhancement mechanism for semantic segmentation and couple it into Swin transformer. Assisting with the GLFI module, the effective modeling of global–local and frequency-domain information can be achieved. The specific network structure is shown in Fig. 1. FGNet uses FGSwin transformer and residual CNNs as the dual-branch encoder to fully extract the features of the image, which contains four stage blocks. In addition, the corresponding features of each stage are fed into GLFI to integrate local and global context expressed by designed FGSwin and CNNs.

B. Frequency-Domain Guided Swin Transformer

Since different information manifests as different frequency components in the frequency domain, the discrimination of some information, including lines and edges, is improved compared with the physical space. Thus, considering enhancing different frequency components to different degrees through learnable weights, self-attention to different frequency components can be achieved, which in turn guides the network to optimize in the direction more conducive to category discrimination. To obtain different frequency components, it is necessary to transform between the physical space and the frequency domain through the fast Fourier transform (FFT) and inverse transform (IFFT) [31].

A single completion of frequency-domain transformation and enhancement is not enough, and an important issue is how to couple the above processes into the deep network structure. Considering different networks have different extraction capabilities, if some information has been neglected in the extraction process, coupling it into the feature fusion does not compensate for the information. Therefore, it becomes one of the feasible ways to couple it into the feature extraction network to guide the process and realize the comprehensive analysis of the frequency domain and physical space information. Swin transformer [17] designs a sliding window mechanism to represent multiscale information by restricting self-attentive computation to nonoverlapping local windows with a low computational complexity, which has been used for semantic segmentation in the past two years and has obtained better result. Hence, a frequency-domain guided Swin (FGSwin) transformer is designed, as shown in Figs. 2 and 3; at the same time, group convolutional structures with different dilation parameters are proposed to obtain the shallow features simultaneously. This processing way can increase the receptive field and reduce the parameters.

Given an image $\mathbf{I}_x$, partitioning it as $\mathbf{I}_{x_i}$ where i is the index of the image patch. Then performing shallow feature extraction using group convolution, which can be represented as

$$\mathbf{x}_i = \mathrm{LN}\left(\mathrm{F}\left(\mathrm{P}\left(\mathbf{I}_{\mathrm{x_i}}\right)\right)\right) \tag{1}$$

$$\mathbf{f}_G = \mathrm{conv}_1\left(\mathrm{CAT}\left(\mathrm{conv}_3^1(\mathbf{x}_\mathrm{i})\mathrm{conv}_1^2(\mathbf{x}_\mathrm{i})\right)\right) \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

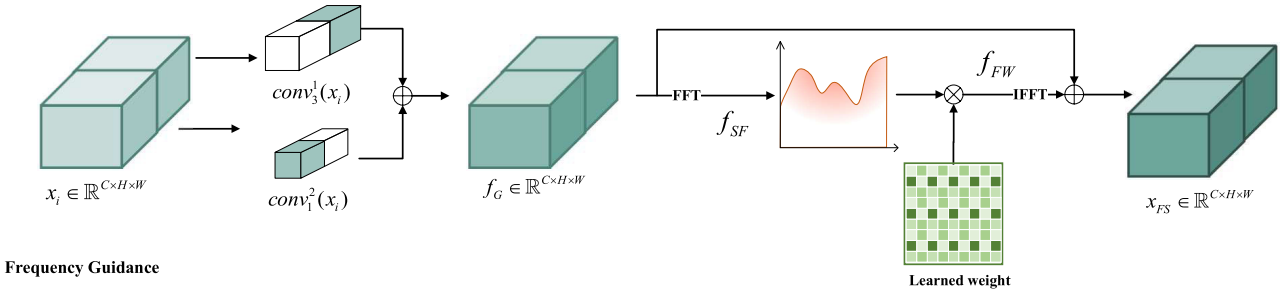


Fig. 2. Details of frequency-domain guidance in FGSwin.

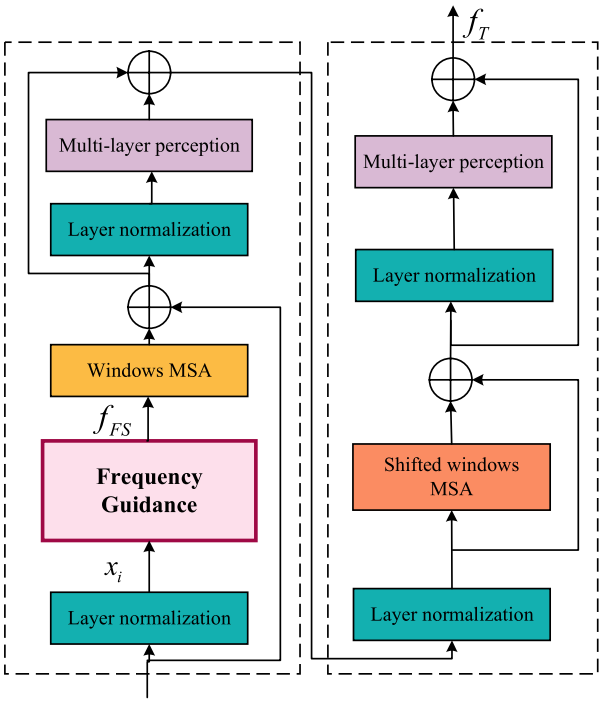


Fig. 3. Architecture of the FGSwin.

where $P(\cdot)$, $F(\cdot)$, and $\mathrm{LN}(\cdot)$ represent the patch embedding, flatten, and layer normalization, respectively. $\mathbf{x}_i$ is the output of the linear embedding and also the input of the frequency guidance. $\mathrm{conv}_{\mathrm{k}}^{\mathrm{d}}(\cdot)$ represents the convolutional operation with the dilation parameter being d and kernel size being k, while $\mathrm{conv}_{\mathrm{k}}(\cdot)$ refers to the convolution operation with kernel size k, and $\mathrm{CAT}(\cdot)$ represents the concatenate operations. $\mathbf{f}_G$ is the output feature of the group convolution and the input feature of FFT. This processing way can reduce the parameter quantity of each convolution layer, thereby preventing overfitting. Furthermore, the use of small-scale convolution kernels and different dilation parameters provide a scheme for developing lightweight models and improving receptive fields [38].

To express the information in the frequency domain, FFT and IFFT are used to transform shallow features in the frequency domain. Furthermore, learnable weights are introduced in the frequency domain to facilitate-adaptive attention to different frequency components, thereby optimizing and enhancing features. The specific calculation formula of the processing is as follows:

$$\mathbf{f}_{\mathrm{SF}} = \mathrm{FFT}(\mathbf{f}_G) \tag{3}$$

$$\mathbf{f}_{\mathrm{FW}} = \mathbf{f}_{\mathrm{SF}} \odot \mathbf{W}_L \tag{4}$$

$$\mathbf{f}_{\mathrm{FS}} = \mathrm{IFFT}(\mathbf{f}_{\mathrm{FW}}) + \mathbf{f}_{\mathrm{G}} \tag{5}$$

where $\mathrm{FFT}(\cdot)$ and $\mathrm{IFFT}(\cdot)$ are the FFT and IFFT operations, respectively, $\mathbf{W}_L$ represents the learnable weight parameter. $\mathbf{f}_{\mathrm{SF}}$ and $\mathbf{f}_{\mathrm{FS}}$ denote the frequency-domain features obtained by FFT and the frequency-domain features obtained by IFFT, respectively. $\mathbf{f}_{\mathrm{FW}}$ is the frequency-domain feature with the learned weight and $\odot$ is the Hadamard product of the two tensors. It can be demonstrated that the frequency-domain enhancement mechanism is not located at the front end of feature coding nor within the fusion module, but rather is coupled within the feature extraction process. Shallow features are often rich in line and edges, and enhancing the frequency domain of shallow features can improve the effectiveness of the input features. Consequently, the guiding mechanism is incorporated into the attention model within the Swin transformer [17] structure to obtain the final features. Specifically, in FGSwin, the output of frequency-domain enhanced feature f_F undergoes layer normalization followed by the windows MSA as follows:

$$\hat{\mathbf{x}}^l = \mathbf{f}_{\mathrm{FS}} + \mathrm{W\text{-}MSA}(\mathrm{LN}(\mathbf{f}_{\mathrm{FS}})) \tag{6}$$

where $\hat{x}^l$ is the output at layer l, $\mathrm{W\text{-}MSA}(\cdot)$ represents the windows MSA, and $\hat{\mathbf{x}}^l$ denotes the output of the windows MSA for block l. This process helps reduce the computational complexity of the traditional self-attention by limiting the scope of attention to smaller regions. Subsequently, the output $\hat{x}^l$ is processed by a multilayer perception after another layer of normalization

$$\mathbf{x}^l = \hat{\mathbf{x}}^l + \mathrm{MLP}\left(\mathrm{LN}\left(\hat{\mathbf{x}}^l\right)\right) \tag{7}$$

where $\mathrm{MLP}(\cdot)$ is the multilayer perception, and $\mathbf{x}^l$ is the output of multilayer perception for block l. This step further refines the representation by incorporating nonlinear transformations while preserving the residual connection for stability. In addition, the output feature is further processed by shift windows MSA, which can be represented as

$$\hat{\mathbf{x}}^{l+1} = \mathbf{x}^l + \mathrm{SW\text{-}WSA}\left(\mathrm{LN}\left(\mathbf{x}^l\right)\right) \tag{8}$$

where $\mathrm{SW\text{-}MSA}(\cdot)$ is the shift windows' MSA, which introduces a shift in the window partitioning process. The output $\hat{x}(l+1)$ is fed into another multilayer perception following normalization, as depicted as

$$\mathbf{x}^{l+1} = \hat{\mathbf{x}}^{l+1} + \mathrm{MLP}\left(\mathrm{LN}\left(\hat{\mathbf{x}}^{l+1}\right)\right). \tag{9}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

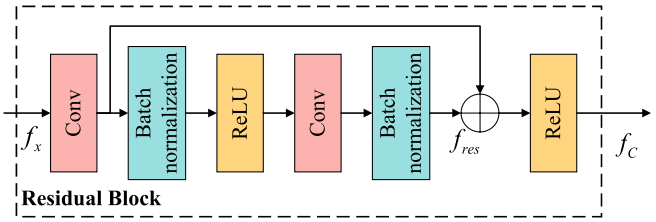


Fig. 4. Structure of the CNNs' branch.

Finally, the features extracted by the FGSwin transformer are fed into the last layer normalization calculated by

$$\mathbf{f}_T = \mathrm{LN}\left(\mathbf{x}^{l+1}\right) \tag{10}$$

where $\mathbf{f}_T$ represents the final output features of the transformer branch. The design of the modules through a multiscale strategy allows for the acquisition of four distinct groups of feature maps, thereby facilitating the effective expression of global–local, frequency-domain, and spatial-domain information.

C. Local Feature Representation by Residual CNNs

FGSwin enables the modeling of long-distance dependence information in the frequency domain and the spatial domain. Although local convolution is introduced in shallow features' encoding, the ability to represent local information of high-level features needs to be enhanced. Therefore, a residual CNN is designed to make up for the above problems. The specific residual block structure is shown in Fig. 4.

Given an input $\mathbf{f}_x$, the residual CNNs' block can be expressed as follows:

$$\mathbf{f}_{\mathrm{res}} = \mathrm{BN}(\mathrm{conv}_1(\mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_3(\mathbf{f}_x))))) \tag{11}$$

$$\mathbf{f}_C = \mathrm{ReLU}((\mathrm{conv}(\mathbf{f}_{\mathrm{x}}) + \mathbf{f}_{\mathrm{res}})) \tag{12}$$

where $\mathrm{BN}(\cdot)$ denotes batch normalization, $\mathrm{ReLU}(\cdot)$ represents the activation function of the rectified linear unit, $\mathbf{f}_{\mathrm{res}}$ is the residual feature, and $\mathbf{f}_C$ is the final feature of the residual CNN branch. The design of the modules through a multiscale strategy allows for the acquisition of four distinct groups of feature maps, thereby facilitating the effective expression of local information.

D. Global and Local Feature Integration and Loss Function

The global and local information of an image can be fully expressed through the designed FGSwin and CNNs. To achieve effective feature fusion, which is different from previous fusion methods using direct stacking or simple connection, GLFI, a lightweight and intuitive feature fusion method, has been designed as Fig. 5.

Assuming that the first stage output features of the FGSwin and residual CNNs are $\mathbf{f}_{T_1}$ and $\mathbf{f}_{C_1}$, respectively, feature fusion can be represented as

$$\mathbf{f}_{u_1} = \mathrm{CAT}\left(\mathbf{f}_{C_1} \odot \mathrm{U}\left(\mathrm{conv}_1\left(\mathbf{f}_{T_1}\right)\right), \mathbf{f}_{C_1}\right) \tag{13}$$

$$\mathbf{f}_1 = \mathrm{SiLU}\left(\mathrm{BN}\left(\mathrm{conv}_3\left(\mathrm{f}_{\mathrm{u}_1}\right)\right)\right) \tag{14}$$

where $\mathbf{f}_{u_1}$ is the first concatenated feature, $\mathrm{U}(\cdot)$ represents upsampling, and $\mathbf{f}_1$ is the first stage fused feature of GLFI. $\mathrm{SiLU}(\cdot)$ is the sigmoid linear unit. This processing way can improve the performance of the model, accelerate the training process, and improve the generalization ability. Similarly, other three fused features $\mathbf{f}_2$, $\mathbf{f}_3$, and $\mathbf{f}_4$ can be obtained. The final four features are concatenated as

$$\mathbf{f} = \mathrm{CAT}(\mathrm{U}(\mathbf{f}_1), \mathrm{U}(\mathbf{f}_2), \mathrm{U}(\mathbf{f}_3), \mathbf{f}_4) \tag{15}$$

where $\mathbf{f}$ is the final feature map.

To achieve the optimal configuration of the entire network structure and obtain the segmentation result, the cross-entropy loss function is applied. At this point, we use the pixelwise representation of the final feature map $\mathbf{f}$, which is the corresponding output from the network, as $\hat{\mathbf{y}}_m^{(n)}$. Thus, the loss function calculated by

$$\mathcal{L}_{ce} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{m=1}^{M}\mathbf{y}_m^{(n)}\log\hat{\mathbf{y}}_m^{(n)} \tag{16}$$

where N and M represent the number of samples and the number of classes, respectively. $\mathbf{y}_m^{(n)}$ is the one-hot vectors of the true labels.

III. Experiments

A. Datasets

To verify the effectiveness of the FGNet, experiments are conducted on two widely used datasets: Potsdam [39] and Vaihingen datasets [40].

1) Potsdam Datasets: The datasets contain 38 high-resolution true orthophoto images of the size of 6000 × 6000, with a ground sampling distance (GSD) of 0.05 m and four channels (red: R, green: G, blue: B, and infrared: IR), which has been annotated with six categories: Car, Tree, Low Vegetation/grass (Low. Veg.), Building, Impervious Surfaces (Imp. Surf.), and Clutter. The IR-R-G false color images are used for the subsequent experiments. Meanwhile, each image is divided into 256 × 256 patches and randomly allocated to the training set and test set, with 16 082 images and 4020 images, respectively.

2) Vaihingen Datasets: The second Vaihingen dataset comprises 33 images, with a GSD of 0.09 m. The largest image in the dataset measures 3816 × 2550 pixels, while the smallest measures 2555 × 1388 pixels. Collectively, these images cover an area of 1.38 square kilometers in Vaihingen, Germany. Each image comprises three bands: IR, R, and G. The dataset has been annotated according to the six categories of the Potsdam. The dataset has been also divided into 256 × 256 patches and randomly split into 3344 train images and 210 test images, respectively.

B. Evaluation Metrics and Implement Details

Three metrics are used to quality the proposed FGNet, including OA, F1-score, and mean intersection over union (mIoU). Among these, OA and F1-score are given as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{17}$$

and

$$\mathrm{F}_1 = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{18}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

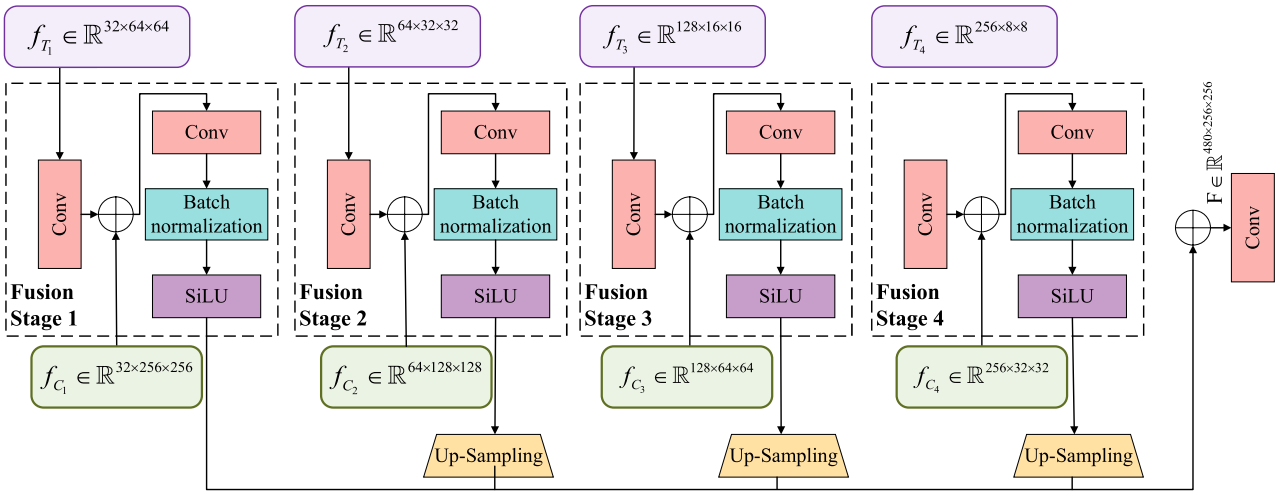


Fig. 5. Structure of the GLFI module.

where OA and F1-score are based on the confusion matrix relying on precision and recall, calculated by

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{19}$$

where TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively. Consequently, for formal consistency, mIoU can be equated with

$$\text{mIoU} = \frac{1}{N}\frac{\text{TP}}{\text{FN}+\text{FP}+\text{TP}} \tag{20}$$

where N is the number of classes.

The mIoU is the average value of intersection over union for all classes, which provides an overall assessment of segmentation accuracy by measuring the degree of overlap between the predicted segmentation mask and the ground-truth mask. The OA is defined as the ratio of the number of correctly predicted pixels to the total number of pixels. Precision and recall are two crucial metrics used to evaluate the performance of models. Precision refers to the proportion of true positive predictions out of all positive predictions made by the model, reflecting ability of the model to avoid false positives. In contrast, recall represents the proportion of true positive predictions out of all actual positive instances, emphasizing the ability to capture all relevant instances and avoid false negatives. A high precision indicates fewer false alarms, while a high recall suggests that the model is successful at identifying most of the relevant instances. However, there is often a tradeoff between precision and recall: improving one can sometimes result in a decrease in the other. This tradeoff is crucial when balancing model performance for different use cases, and the F1-score provides a harmonic mean of both to offer a balanced measure when both the metrics are important. It can be demonstrated that the greater the values of the aforementioned indicators, the more accurate the segmentation result.

The experimental environment used a server equipped with an 13th Gen Intel[1] Core[2] i9-13900K CPU and an NVIDIA GeForce RTX 4090 24G GPU running Linux version Ubuntu 20.04.4LTS. All the comparison methods use their public code and are conducted within the same test framework and environment. The optimal parameters are selected as well. Adam optimizer is used with an initial learning rate of 0.001 and a weight decay of $2.5e^{-4}$. To manage the learning rate, we use the cosine annealing strategy with a warmup and restart, whose hyperparameters are the number of iterations for the first restart and the factor increases of the restart, which are 15 and 2, respectively. In addition, we set a batch size of 8 and 300 epochs.

[1]Registered trademark.
[2]Trademarked.

C. Comparison Methods

A comprehensive array of benchmark methods was selected for quantitative comparison, including FCN [41], U-net [42], SwinB-UperNet [17], SegFormer [13], UNetFormer [18], SRCBTFusion [21], STUnet [22], CMTFNet [19], MsanlfNet [31], SFFNet [28], and MIFNet [32].

1) FCN: As the one of the most classic CNNs' method, FCN uses the fully connected layer to replace with a convolution layer, enabling pixelwise classification.

2) U-Net: U-net represents a further SOTA method of the most classic CNNs, which is structured as a symmetric encoder–decoder comprising dual paths for feature extraction and information propagation.

3) SwinB-UperNet: It is an outperforming transformer architecture combining FPN and Swin transformer to represent global prior information and reduce computational burden.

4) SegFormer: SegFormer is an SOTA transformer architecture model, which comprises a hierarchically structured transformer encoder that outputs multiscale features.

5) UNetFormer: UNetFormer is a hybrid model combining CNNs and transformer, using ResNet as the encoder and a transformer-based decoder to capture both global and local information effectively.

6) SRCBTFusion: It adopts a cascade CNN–transformer encoder–decoder structure with semantic information and edge segmentation to improve the feature selection and feature aggregation.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

7) STUNet: STUNet combines the Swin transformer with CNNs in a dual-encoder structure; meanwhile, it can enhance the feature representation and accuracy through spatial interaction, feature compression, and relational aggregation.

8) CMTFNet: CMTFNet, as a CNN–transformer hybrid model, integrates local information from CNNs with multiscale global information from transformer. Moreover, it uses a multiscale MSA and the attention fusion module to enhance feature extraction.

9) MsanlfNet: This method uses the combination of feature fusion module to achieve frequency-domain enhancement, thereby attaining a superior segmentation effect. The method is selected for analysis to ascertain the advantages of double branches and to verify the advantages of coupling frequency guidance into the processing of feature extraction.

10) SFFNet: SFFNet introduces the Haar-wavelet-based wavelet transform feature decomposer, which separates and integrates both low- and high-frequency components with spatial features.

11) MIFNet: MIFNet builds on this by incorporating local, global, and frequency data within a unified module, which adaptively associates these dimensions.

D. Comparison to SOTA Methods on Potsdam Datasets

The specific experimental results on the Potsdam Dataset are shown in Table I and Fig. 6. It can be observed that methods based on CNNs or transformers alone achieve relatively low accuracy, primarily because relying on either a CNN or transformer feature extraction framework independently is insufficient for comprehensive information extraction at both global and local levels. For example, FCN achieves 67.49% mIoU and U-net achieves 68.71% mIoU. They perform well in capturing local spatial features through its convolutional layers, yet struggle to effectively capture global context, which is essential for accurately segmenting complex remote sensing images. SwinB-UperNet and Segformer achieve 65.88% and 64.67% mIoU, which introduce transformer-based mechanisms to capture global information, but lack the ability to model fine-grained frequency details that are critical for delineating precise boundaries and small-scale structures. The integration of CNNs and transformer methods, such as UNetFormer and SRCBTFusion, has led to an enhancement in the performance of semantic segmentation. This is evidenced by the improved mIoU of 68.78% and 69.52%, notably within the category of "Imp. Surf" and "building." It is reasonably concluded that this comprehensive hybrid architecture is capable of enhancing the distinction between classes and consequently improving the segmentation effect. To facilitate a more detailed comparison of the impact of different hybrid methods, STUNet and CMTFNet are replicated and achieve corresponding improvements. While the aforementioned four methods (UNetFormer, SRCBTFusion, STUnet, and CMTFNet) demonstrate enhanced segmentation outcomes of hybrid architecture, the segmentation results of STUNet and CMTFNet using the two-branch stream are more optimal, reaching 69.67% and 71.21% mIoU, respectively. In particular, CMTFNet demonstrates the most effective segmentation performance in the categories of "Building," "Low. Veg.," and "Car," with the respective mIoU of 86.52%, 70.07%, and 75.07%. To illustrate the advantages of frequency-domain guidance, a comparison between MsanlfNet, SFFNet, MIFNet, and the proposed FGNet is presented. The results demonstrate that the mIoU for the three methods compared is 70.95%, 65.47%, and 70.36%, respectively, all of which are inferior to the performance of the proposed method. This superiority can be attributed to the novel integration of frequency-domain information within our approach. By simultaneously modeling both the spatial and frequency domain features, our method significantly enhances the feature representation capabilities for remote sensing images. In contrast, the coupling mechanisms used by the other methods fall short, particularly in effectively distinguishing between various feature types. This underscores the advantages of our method in leveraging frequency-domain information and highlights its superior effectiveness compared with alternative coupling strategies.

Moreover, the proposed FGNet has been shown to perform the best segmentation result through a qualitative analysis of multiple images. As illustrated in Fig. 6, the yellow box represents the most significant improvement of the proposed method. It is obvious that FGNet exhibits a distinct boundary, high class consistency, and an effective segmentation result.

E. Comparison to SOTA Methods on Vaihingen Datasets

The experimental results on the Vaihingen datasets are shown in Table II and Fig. 7. Similar to the findings on Potsdam dataset, it is observed that methods alone based on CNNs or transformers exhibit relatively low accuracy. UNetFormer and SRCBTFusion enhance segmentation performance with the cascade hybrid architecture, achieving mIoU of 71.76% and 72.98%, respectively. Furthermore, the two-branch hybrid architectures CMTFNet and STUNet also show better results than the cascade architecture and obtain more optimal segmentation results, with mIoU reaching 70.45% and 73.21%. Specifically, CMTFNet demonstrates superior segmentation performance in the categories of "Imp. Surf," "Low. Veg.," and "Tree" is improved to 77.78%, 59.48%, and 73.21%, respectively. Similarly, MsanlfNet, SFFNet, and MIFNet, which incorporate frequency-domain guidance within CNN methods, enhance segmentation performance, with achieving an mIoU of 73.21%, 70.12%, and 72.99%, respectively. The proposed FGNet achieves the highest segmentation performance again. The frequency guidance mechanism and the feature fusion module optimize feature quality, leading to superior segmentation results. It has also been demonstrated that the utilization of the frequency domain can enhance the results of semantic segmentation. Moreover, the proposed FGNet achieves the highest segmentation performance again. The frequency guidance mechanism and the feature fusion module optimize feature quality, leading to superior segmentation results. The comparative analysis between MsanlfNet and FGNet confirms that the proposed coupling approach is more effective, highlighting the efficacy of the designed coupling method in improving feature effectiveness and class separability.

Similarly, through qualitative analysis as shown in Fig. 7, the proposed method FGNet also achieves the best segmentation performance on the Vaihingen dataset. The yellow

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I

COMPARISON TO SOTA METHODS ON THE POTSDAM DATASET. THE BOLD FONT INDICATES THE BEST DATA, WHILE THE UNDERLINED DENOTES THE SECOND-BEST DATA

Methods	IoU/%						mIoU/%	OA/%	F1/%	Precision/%	Recall/%
	Imp. Suf.	Building	Low. Veg.	Tree	Car	Clutter					
FCN [41]	75.88	84.22	65.48	64.84	72.13	42.37	67.49	83.70	79.81	80.01	79.88
U-net [42]	76.73	84.81	67.52	66.89	73.50	42.81	68.71	84.49	80.68	82.00	79.74
SwinB-UperNet [17]	73.87	81.48	65.64	62.25	69.36	42.70	65.88	82.57	78.75	79.39	78.44
SegFormer [13]	73.47	80.88	63.69	60.86	69.04	40.07	64.67	81.74	77.75	78.01	77.66
UNetFormer [18]	77.42	84.98	67.05	64.55	73.81	44.90	68.78	84.44	80.80	81.92	80.04
SRCBTFusion [21]	78.11	85.96	68.73	65.62	71.73	46.98	69.52	85.24	81.59	81.50	81.79
STUNet [22]	77.64	86.14	68.10	66.63	72.47	47.03	69.67	84.24	80.67	80.84	80.66
CMTFNet [19]	78.74	86.52	70.07	**68.07**	75.07	48.78	71.21	85.95	82.60	**83.97**	81.51
MsanlfNet [31]	78.94	**87.34**	69.50	67.28	74.19	48.44	70.95	84.86	82.37	83.26	81.61
SFFNet [28]	74.45	81.49	64.61	59.24	69.88	43.16	65.47	82.26	78.44	79.77	77.81
MIFNet [32]	78.66	86.60	69.42	66.22	73.85	47.42	70.36	85.58	81.97	83.03	81.19
Ours (FGNet)	**80.09**	85.93	**70.16**	67.81	**75.64**	**49.14**	**71.46**	**86.10**	**82.78**	83.63	**82.09**

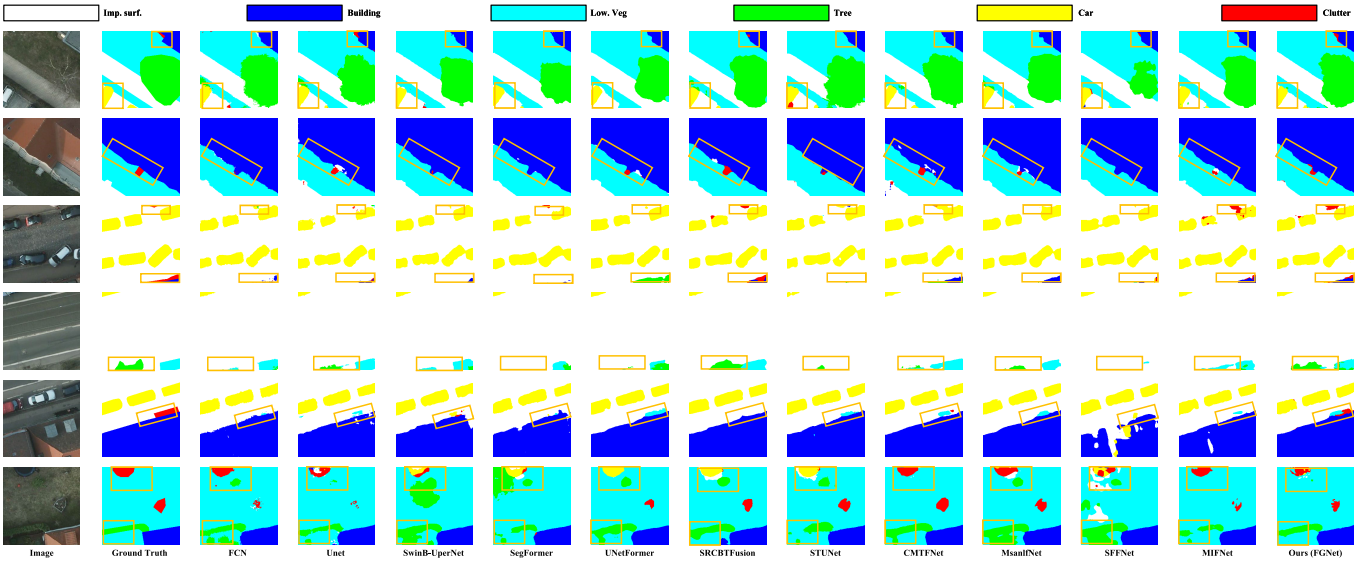


Fig. 6. Visualization results of the Potsdam datasets.

TABLE II

COMPARISON TO SOTA METHODS ON THE VAIHINGEN DATASET. THE BOLD INDICATES THE BEST DATA, WHILE THE UNDERLINED DENOTES THE SECOND-BEST DATA

Methods	IoU/%						mIoU/%	OA/%	F1/%	Precision/%	Recall/%
	Imp. Suf.	Building	Low. Veg.	Tree	Car	Clutter					
FCN [41]	77.74	84.23	58.99	71.87	49.91	84.54	71.21	85.04	82.49	84.13	81.12
U-net [42]	75.43	83.65	55.70	71.62	57.11	83.33	71.14	85.10	82.74	84.68	81.45
SwinB-UperNet [17]	77.21	82.65	59.44	**73.74**	50.22	86.74	71.66	85.16	82.81	84.65	81.28
SegFormer [13]	77.20	82.81	58.30	72.25	48.04	89.21	71.29	84.76	82.41	84.87	80.58
UnetFormer [18]	77.23	82.48	57.87	70.01	51.12	**91.82**	71.76	84.30	82.77	85.07	81.00
SRCBFusion [21]	77.24	83.99	58.27	72.53	55.80	90.07	72.98	85.15	83.76	85.06	82.70
STUNet [22]	75.11	81.64	56.39	70.36	49.72	89.51	70.45	83.58	81.88	84.29	79.96
CMTFNet [19]	77.78	83.97	59.48	72.52	52.17	89.60	72.59	85.31	83.42	85.22	82.04
MsanlfNet [31]	**79.63**	**85.36**	**60.72**	73.21	55.10	85.24	73.21	**86.06**	83.99	84.74	83.44
SFFNet [28]	76.50	82.50	57.75	71.05	44.48	88.43	70.12	84.22	81.46	83.11	80.17
MIFNet [32]	78.67	84.33	59.35	73.21	56.94	84.52	72.99	85.17	83.75	85.33	82.55
Ours (FGNet)	77.67	84.35	59.58	73.51	**57.39**	91.73	**74.04**	85.62	**84.49**	**85.65**	**83.51**

boxes highlight the most significant improvements. The FGNet demonstrates superior segmentation results with distinct boundaries and high class consistency.

A further examination of the model efficiency reveals that the proposed method exhibits the lowest for parameters (params) and the floating point of operations (FLOPs), as is shown in Table III. The rationale behind this lies in the fact that the effective parameters of the CNNs' branch structure are minimal, and the incorporation of group convolution further reduces the parameters of the FGSwin. In addition, GLFI is characterized by a minimalist approach, with small parameters and a simple structure. The aforementioned design elements contribute to the overall lightweight construction of the FGNet.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

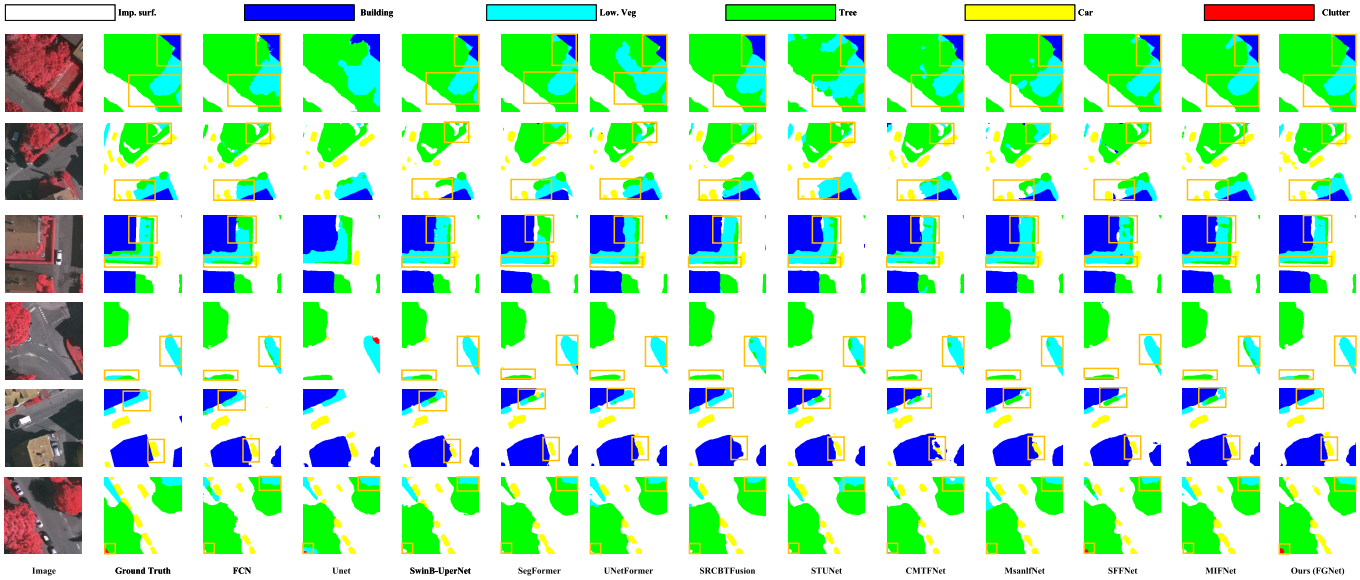


Fig. 7. Visualization results of the Vaihingen datasets.

TABLE III
COMPARISON TO SOTA METHODS OF PARAMS AND FLOPS

Methods	Parameters/M	FLOPS/G
FCN [41]	18.64	430.12
U-net [42]	13.40	249.12
SwinB-UperNet [17]	33.24	430.12
SegFormer [13]	26.52	32.24
UNetFormer [18]	11.68	23.48
SRCBTFusion [21]	74.41	256.63
STUNet [22]	325.35	157.50
CMTFNet [19]	30.07	68.56
MsanlfNet [31]	32.24	55.62
SFFNet [28]	103.35	34.16
MIFNet [32]	70.84	25.35
Ours (FGNet)	**8.08**	**8.02**

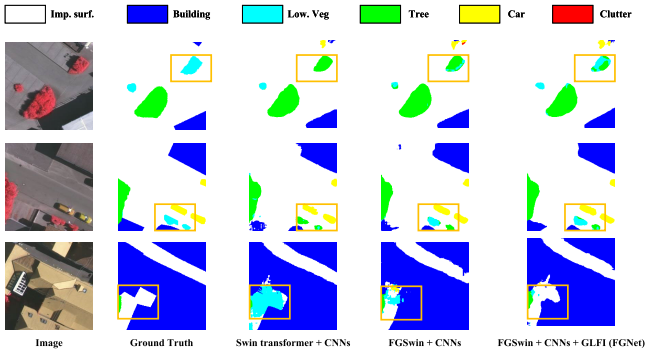


Fig. 8. Visualization results of ablation study.

F. Ablation Study

Take Vaihingen dataset as example, in this part, ablation study of the proposed method is constructed. The specifically experimental results are shown in Table IV and Fig. 8. In these experiments, "Swin + CNNs" refers to a dual-branch structure where the upper branch uses the Swin transformer as the feature extraction model, and the lower branch uses the designed residual CNN model. The segmentation results are obtained by directly using the upsampling and stacking approach. The "FGSwin + CNNs" method couples the frequency-domain guidance mechanism to the feature extraction of the Swin converter under the same dual-branch architecture. This configuration demonstrates the impact of the devised frequency-domain guidance mechanism. The "FGSwin + CNNs + GLFI" denotes the aforementioned cube with the GLFI module, which is subjected to further analysis to ascertain its efficiency.

Through the quantitative results in Table IV and the segmentation results in Fig. 8, it can be observed that the method using only "Swin transformer + CNNs" achieved relatively low segmentation performance, with mIoU, OA, and F1-score of 70.41%, 87.50%, and 69.69%, respectively. The proposed frequency-domain guidance mechanism, coupling strategy, and GLFI module demonstrate significant advantages and effectiveness. Specifically, "FGSwin + CNNs" obtains 72.95% mIoU, 94.04% OA, and 76.23% F1-score, which has an obvious increasing of about 2.5% mIoU, 6.5% OA, and 6.6% F1-score. We contend that the process of feature extraction is enhanced by the application of frequency-domain guidance. Furthermore, the "FGSwin + CNNs + GLFI" approach results in an additional increase of over 1% in mIoU, reaching a total of 74.04%. The GLFI module is also capable of enhancing classification separability through the utilization of a small-scale convolutional kernel and SiLU. In addition, through qualitative analysis as shown in Fig. 8, the yellow boxes highlight the most significant improvements. The proposed FGNet exhibits superior segmentation results with higher class consistency and discernibility. These collectively contributed to a notable enhancement in performance, as evidenced by the quantitative and qualitative visual analyses.

IV. DISCUSSION

To further analyze the performance of our proposed method, we visualize the feature maps of the FGSwin transformer alongside the baseline Swin transformer, as shown in Fig. 9. The visualization enables an in-depth exploration of the mechanisms that enhance segmentation accuracy, especially

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
ABLATION EXPERIMENTS OF THE PROPOSED MODULE ON THE VAIHINGEN DATASET. THE BOLD INDICATES THE BEST DATA

Methods	IoU/%						mIoU/%	OA/%	F1/%
	Imp. Suf.	Building	Low. Veg.	Tree	Car	Clutter			
Swin + CNNs	76.80	83.24	59.42	**73.97**	56.09	72.97	70.41	85.07	82.26
FGSwin + CNNs	**77.81**	83.74	58.18	72.39	55.72	89.87	72.95	85.11	83.74
FGSwin + CNNs + GLFI (FGNet)	77.67	**84.35**	**59.58**	73.51	**57.39**	**91.73**	**74.04**	**85.62**	**84.49**

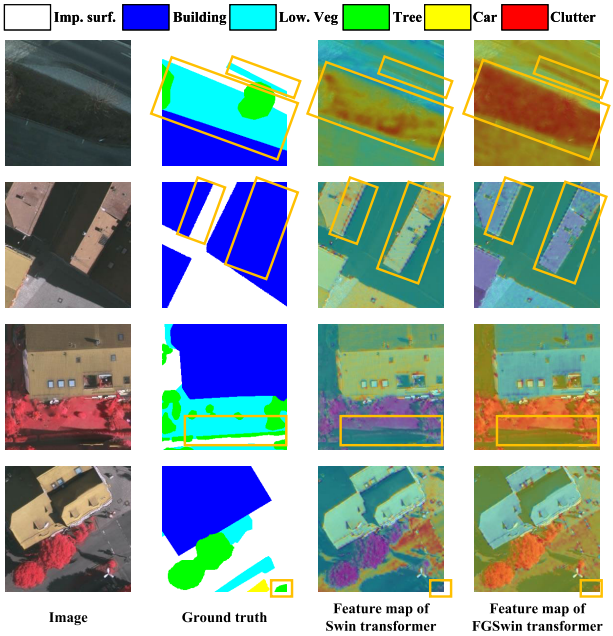


Fig. 9. Visualization of mid-feature maps of Swin transformer and FGSwin transformer.

in contexts involving edge and the analysis of homogeneity and heterogeneity within images.

The results demonstrate that the frequency-domain guided FGSwin transformer significantly enhances boundary sharpness and preserves intricate details. In contrast, the Swin transformer feature maps exhibit a lower level of clarity, suggesting challenges in capturing fine-grained details without explicit frequency-based enhancements. This limitation may result in inaccuracies, especially in areas with complex textures or abrupt transitions between classes. For example, in the first row, the FGSwin transformer demonstrates enhanced intraclass consistency within the central "Low Veg." category, effectively maintaining uniformity across this region. These improvements underscore the ability of proposed FGSwin to leverage frequency-domain information for superior edge preservation and more cohesive feature representation across complex textures. It enhances the ability of our network to capture and integrate frequency-domain information during the feature extraction process. This enhancement facilitates improved handling of both global and local details, which demonstrates the advantages of our frequency-guided approach in complex remote sensing scenarios.

V. CONCLUSION

This article proposes a lightweight semantic segmentation framework based on a dual-branch hybrid architecture. By incorporating a frequency-domain enhancement mechanism into the feature extraction framework, it is possible to ensure comprehensive expression of global–local, frequency-domain, and spatial-domain information. Furthermore, a residual convolution network and a feature fusion module using small-scale convolution kernels have been devised, which serve to enhance the capacity for semantic feature extraction and improve the classification separability and segmentation effect. Two publicly available datasets, Potsdam and Vaihingen, are used in the experiments, and a comparison of the current SOTA semantic segmentation methods demonstrates that the proposed method exhibits excellent segmentation performance.

REFERENCES

[1] Z. Cai et al., "Improving agricultural field parcel delineation with a dual branch spatiotemporal fusion network by integrating multimodal satellite data," *ISPRS J. Photogramm. Remote Sens.*, vol. 205, pp. 34–49, Nov. 2023.

[2] X. Huang, W. Wang, J. Li, L. Wang, and X. Xie, "A stepwise refining image-level weakly supervised semantic segmentation method for detecting exposed surface for buildings (ESB) from very high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5400517.

[3] E. Zhang, H. Zong, X. Li, M. Feng, and J. Ren, "ICSF: Integrating inter-modal and cross-modal learning framework for self-supervised heterogeneous change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5501516.

[4] Y. He, J. Wang, Y. Zhang, and C. Liao, "An efficient urban flood mapping framework towards disaster response driven by weakly supervised semantic segmentation with decoupled training samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 207, pp. 338–358, Jan. 2024.

[5] M. Pastorino, G. Moser, S. B. Serpico, and J. Zerubia, "Semantic segmentation of remote-sensing images through fully convolutional neural networks and hierarchical probabilistic graphical models," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5407116.

[6] L. Liu et al., "HierU-Net: A hierarchical semantic segmentation method for land cover mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4404614.

[7] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[8] A. Kirillov, R. Girshick, K. He, and P. Dollár, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6392–6401.

[9] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4095–4104.

[10] J. Wang, Z. Feng, Y. Jiang, S. Yang, and H. Meng, "Orientation attention network for semantic segmentation of remote sensing images," *Knowledge-Based Syst.*, vol. 267, May 2023, Art. no. 110415.

[11] J. Liu, W. Hua, W. Zhang, F. Liu, and L. Xiao, "Stair fusion network with context-refined attention for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701517.

[12] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, Jan. 2020, pp. 1–21.

[13] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.

[14] R. Xu, C. Wang, J. Zhang, S. Xu, W. Meng, and X. Zhang, "RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 1052–1064, 2023.
[15] C. You et al., "Boundary-aware multiscale learning perception for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4407115.
[16] Z. Xu, J. Geng, and W. Jiang, "MMT: Mixed-mask transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5613415.
[17] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[18] L. Wang et al., "Unetformer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.
[19] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.
[20] S. Du and M. Liu, "Class-guidance network based on the pyramid vision transformer for efficient semantic segmentation of high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5578–5589, 2023.
[21] J. Chen, J. Yi, A. Chen, and H. Lin, "SRCBTFusion-Net: An efficient fusion architecture via stacked residual convolution blocks and transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4411716.
[22] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.
[23] H. Wang, X. Li, L. Huo, and C. Hu, "Global and edge enhanced transformer for semantic segmentation of remote sensing," *Int. J. Speech Technol.*, vol. 54, no. 7, pp. 5658–5673, Apr. 2024.
[24] L. Fan, Y. Zhou, H. Liu, Y. Li, and D. Cao, "Combining Swin transformer with UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530111.
[25] L. Chen, L. Gu, D. Zheng, and Y. Fu, "Frequency-adaptive dilated convolution for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 3414–3425.
[26] X. Li, F. Xu, H. Gao, F. Liu, and X. Lyu, "A frequency domain feature-guided network for semantic segmentation of remote sensing images," *IEEE Signal Process. Lett.*, vol. 31, pp. 1369–1373, 2024.
[27] J. Liu, D. Zhang, L. He, X. Yu, and W. Han, "MFAGNet: Multiscale frequency attention gating network for land cover classification," *Int. J. Remote Sens.*, vol. 44, no. 21, pp. 6670–6697, Nov. 2023.
[28] Y. Yang, G. Yuan, and J. Li, "SFFNet: A wavelet-based spatial and frequency domain fusion network for remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3000617.
[29] C. Yang and Z. Zhang, "PFD-net: Pyramid Fourier deformable network for medical image segmentation," *Comput. Biol. Med.*, vol. 172, Mar. 2024, Art. no. 108302.
[30] X. Li et al., "SSCNet: A spectrum-space collaborative network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 15, no. 23, p. 5610, Dec. 2023.
[31] L. Bai, X. Lin, Z. Ye, D. Xue, C. Yao, and M. Hui, "MsanlfNet: Semantic segmentation network with multiscale attention and nonlocal filters for high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[32] J. Fan, J. Li, Y. Liu, and F. Zhang, "Frequency-aware robust multidimensional information fusion framework for remote sensing image segmentation," *Eng. Appl. Artif. Intell.*, vol. 129, Mar. 2024, Art. no. 107638.
[33] G. Wei, J. Xu, W. Yan, Q. Chong, H. Xing, and M. Ni, "Dual-domain fusion network based on wavelet frequency decomposition and fuzzy spatial constraint for remote sensing image segmentation," *Remote Sens.*, vol. 16, no. 19, p. 3594, Sep. 2024.
[34] Y. Li, Z. Liu, J. Yang, and H. Zhang, "Wavelet transform feature enhancement for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 15, no. 24, p. 5644, Dec. 2023.
[35] T. Zhang and R. P. Dick, "Spatial-frequency network for segmentation of remote sensing images," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2023, pp. 3553–3557.
[36] X. Zhou et al., "MeSAM: Multiscale enhanced segment anything model for optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5623515.
[37] J. Li, S. Zhang, Y. Sun, Q. Han, Y. Sun, and Y. Wang, "Frequency-driven edge guidance network for semantic segmentation of remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9677–9693, 2024.
[38] F. Yu and V. Koltun, "Multi-scale context aggregation by dilated convolutions," in *Proc. 4th Int. Conf. Learn. Represent. (ICLR)*, San Juan, Puerto Rico, Y. Bengio and Y. LeCun, Eds., May 2016, pp. 1–13.
[39] L. Fang, P. Zhou, X. Liu, P. Ghamisi, and S. Chen, "Context enhancing representation for semantic segmentation in remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 4138–4152, Mar. 2024.
[40] X. Li, F. Xu, F. Liu, Y. Tong, X. Lyu, and J. Zhou, "Semantic segmentation of remote sensing images by interactive representation refinement and geometric prior-guided inference," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5400318.
[41] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[42] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, N. Navab, J. Hornegger, W. M. Wells, and A. F. Frangi, Eds., Cham, Switzerland: Springer, 2015, pp. 234–241.

Haoxue Zhang received the B.Eng. degree from Beijing University of Chemical Technology, Beijing, China, in 2017. She is currently pursuing the Ph.D. degree in control science and engineering with the School of Electronic and Information Engineering, Taiyuan University of Science and Technology, Taiyuan, China.

Her research interests include scene understanding, semantic segmentation, and high-resolution remote sensing image processing.

Gang Xie (Member, IEEE) received the B.S. degree in control theory and the Ph.D. degree in circuits and systems from Taiyuan University of Technology, Taiyuan, China, in 1994 and 2006, respectively.

He has been the Vice President of the Taiyuan University of Science and Technology, Taiyuan, and a Professor with the Taiyuan University of Technology, since 2008. His research interests include intelligent information processing, complex networks, and big data.

Linjuan Li received the B.S. degree from Taiyuan University of Technology, Taiyuan, China, in 2009, and the Ph.D. degree in control science and engineering from Taiyuan University of Science and Technology, Taiyuan, in 2024.

She is currently with the School of Electronic Information Engineering, Taiyuan University of Science and Technology. Her research interests include computer vision and deep learning technology.

Xinlin Xie received the B.S. degree in electronics and information engineering and the Ph.D. degree in electronic science and technology from Taiyuan University of Technology, Taiyuan, China, in 2012 and 2019, respectively.

He is currently with the School of Electronic Information Engineering, Taiyuan University of Science and Technology, Taiyuan. His research interests include deep learning technology and computer vision.

Jinchang Ren (Senior Member, IEEE) received the B.Eng., M.Eng., and D.Eng. degrees from Northwestern Polytechnical University, Xi'an, China, in 1992, 1997, and 2000, respectively, and the Ph.D. degree from the University of Bradford, Bradford, U.K., in 2019.

He is currently with the School of Computer Science, Guangdong Polytechnic Normal University, Guangzhou, China. He is also with the School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, U.K. His research interests include hyperspectral imaging, image processing, computer vision, big data analytics, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 21:04:48 UTC from IEEE Xplore. Restrictions apply.